

Linear Algebra on a Computer

An Introduction to Black Box Methods

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Outline

Linear Algebra
on a
Computer

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Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

- 1 Symbolic Computation
- 2 Symbolic Algorithms
- 3 Black Box Matrix Model
- 4 Wiedemann Method
 - Background
 - The Method

Symbolic Computation

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Superset of computer algebra

Symbolic Computation

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Superset of computer algebra

- Symbols or exact arithmetic

Symbolic Computation

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Superset of computer algebra

- Symbols or exact arithmetic
- Based on exact finite representation of finite or infinite mathematical objects

Symbolic Computation

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Superset of computer algebra

- Symbols or exact arithmetic
- Based on exact finite representation of finite or infinite mathematical objects
- Abstract mathematical structures (groups, rings, fields, etc.)

Computar Algebra Systems

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

General Purpose Systems

Computar Algebra Systems

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

General Purpose Systems

- AXIOM, MAGMA, Maple, *Mathematica*, REDUCE

Computar Algebra Systems

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

General Purpose Systems

- AXIOM, MAGMA, Maple, *Mathematica*, REDUCE

Special Purpose Systems

Computar Algebra Systems

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

General Purpose Systems

- AXIOM, MAGMA, Maple, *Mathematica*, REDUCE

Special Purpose Systems

- CoCoA (Computations in Commutative Algebra)

Computar Algebra Systems

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

General Purpose Systems

- AXIOM, MAGMA, Maple, *Mathematica*, REDUCE

Special Purpose Systems

- CoCoA (Computations in Commutative Algebra)
- GAP (Groups, Algorithms, and Programming)

Computar Algebra Systems

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

General Purpose Systems

- AXIOM, MAGMA, Maple, *Mathematica*, REDUCE

Special Purpose Systems

- CoCoA (Computations in Commutative Algebra)
- GAP (Groups, Algorithms, and Programming)
- NTL (Number Theory Library)

Computar Algebra Systems

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

General Purpose Systems

- AXIOM, MAGMA, Maple, *Mathematica*, REDUCE

Special Purpose Systems

- CoCoA (Computations in Commutative Algebra)
- GAP (Groups, Algorithms, and Programming)
- NTL (Number Theory Library)
- SINGULAR (polynomial computations)

Long History

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Ancient Algorithms

Long History

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Ancient Algorithms

- Euclidean algorithm for finding the greatest common divisor

Long History

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Ancient Algorithms

- Euclidean algorithm for finding the greatest common divisor
- Chinese remainder algorithm

Long History

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Ancient Algorithms

- Euclidean algorithm for finding the greatest common divisor
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Isaac Newton's *The Universal Arithmetic* (1728)

Systematically discusses rules for manipulating universal mathematical expressions, that is, formulae containing symbolic indeterminates, and algorithms for solving equations built with these expressions.

Symbolic Computation vs. Numerical Analysis

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Numerical Analysis

Symbolic Computation vs. Numerical Analysis

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Numerical Analysis

- Floating point numbers (approximate real values)

Symbolic Computation vs. Numerical Analysis

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Numerical Analysis

- Floating point numbers (approximate real values)
- Find approximation quickly

Symbolic Computation vs. Numerical Analysis

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Numerical Analysis

- Floating point numbers (approximate real values)
- Find approximation quickly
- Error propagation important

Symbolic Computation vs. Numerical Analysis

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Numerical Analysis

- Floating point numbers (approximate real values)
- Find approximation quickly
- Error propagation important

Condition Numbers: amplification factors of relative errors

Symbolic Computation vs. Numerical Analysis

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Numerical Analysis

- Floating point numbers (approximate real values)
- Find approximation quickly
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Condition Numbers: amplification factors of relative errors

Stability: whether all roundoff errors of an algorithm
are harmless

Symbolic Computation vs. Numerical Analysis

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Numerical Analysis

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are harmless

Symbolic Computation

Symbolic Computation vs. Numerical Analysis

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Numerical Analysis

- Floating point numbers (approximate real values)
- Find approximation quickly
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Condition Numbers: amplification factors of relative errors

Stability: whether all roundoff errors of an algorithm
are harmless

Symbolic Computation

- Find exact solution quickly

Symbolic Computation vs. Numerical Analysis

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Numerical Analysis

- Floating point numbers (approximate real values)
- Find approximation quickly
- Error propagation important

Condition Numbers: amplification factors of relative errors

Stability: whether all roundoff errors of an algorithm
are harmless

Symbolic Computation

- Find exact solution quickly
- May never approximate (may have no metric)

Symbolic Computation vs. Numerical Analysis

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Algorithms may not be compatible

Symbolic Computation vs. Numerical Analysis

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Algorithms may not be compatible

- Numerical algorithms may never find exact solution

Symbolic Computation vs. Numerical Analysis

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Algorithms may not be compatible

- Numerical algorithms may never find exact solution
- Symbolic algorithms may be ill conditioned or unstable

Symbolic Computation vs. Numerical Analysis

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

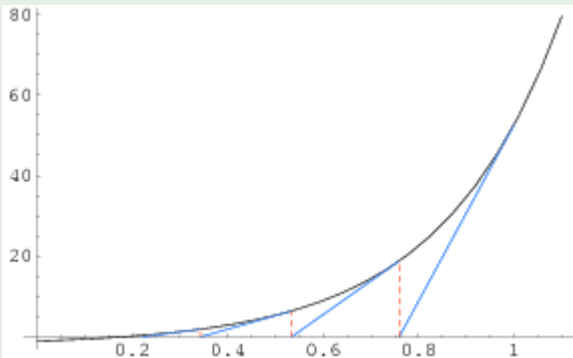
Wiedemann
Method

Background
The Method

Algorithms may not be compatible

- Numerical algorithms may never find exact solution
- Symbolic algorithms may be ill conditioned or unstable

Example (Newton's Method)



Infinite Mathematical Objects

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

General Approach

Infinite Mathematical Objects

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

General Approach

- Computer has finite memory

Infinite Mathematical Objects

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

General Approach

- Computer has finite memory
- Cannot compute exactly over reals, rationals, integers, etc.

Infinite Mathematical Objects

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

General Approach

- Computer has finite memory
- Cannot compute exactly over reals, rationals, integers, etc.
- Compute bound on desired solution

Infinite Mathematical Objects

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

General Approach

- Computer has finite memory
- Cannot compute exactly over reals, rationals, integers, etc.
- Compute bound on desired solution
- Use finite field methods to find solution modulo p_i

Infinite Mathematical Objects

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

General Approach

- Computer has finite memory
- Cannot compute exactly over reals, rationals, integers, etc.
- Compute bound on desired solution
- Use finite field methods to find solution modulo p_i
- Reconstruct solution (e.g., Chinese remainder algorithm)

Probabilistic Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo Algorithm

Probabilistic Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo Algorithm

- Always fast; Probably correct

Probabilistic Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo Algorithm

- Always fast; Probably correct

Las Vegas Algorithm

Probabilistic Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo Algorithm

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Las Vegas Algorithm

- Probably fast; Always correct

Probabilistic Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo Algorithm

- Always fast; Probably correct

Las Vegas Algorithm

- Probably fast; Always correct

Bounded-error, Probabilistic, Polynomial time (BPP)

Probabilistic Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo Algorithm

- Always fast; Probably correct

Las Vegas Algorithm

- Probably fast; Always correct

Bounded-error, Probabilistic, Polynomial time (BPP)

- Probably fast; Probably correct

Probabilistic Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo Algorithm

- Always fast; Probably correct

Las Vegas Algorithm

- Probably fast; Always correct

Bounded-error, Probabilistic, Polynomial time (BPP)

- Probably fast; Probably correct
- Atlantic City?

Probabilistic Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Schwartz-Zippel Lemma

Let R be an integral domain, $n \in \mathbb{N}$, $S \subset R$ finite with $s = |S|$ elements, and $f \in R[\lambda_1, \dots, \lambda_n]$ a polynomial of total degree at most $d \in \mathbb{N}$.

- 1 If f is not the zero polynomial, then f has at most ds^{n-1} zeros in S^n .
- 2 If $s > d$ and f vanishes on S^n , then $f = 0$.

Converting Algorithms

Monte Carlo to Las Vegas

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Converting Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo to Las Vegas

- Requires certificate to check if solution is correct

Converting Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo to Las Vegas

- Requires certificate to check if solution is correct
- If not correct, run Monte Carlo Algorithm again

Converting Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo to Las Vegas

- Requires certificate to check if solution is correct
- If not correct, run Monte Carlo Algorithm again

Las Vegas to Monte Carlo

Converting Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo to Las Vegas

- Requires certificate to check if solution is correct
- If not correct, run Monte Carlo Algorithm again

Las Vegas to Monte Carlo

- Start Las Vegas algorithm and let run for set period of time

Converting Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo to Las Vegas

- Requires certificate to check if solution is correct
- If not correct, run Monte Carlo Algorithm again

Las Vegas to Monte Carlo

- Start Las Vegas algorithm and let run for set period of time
- Will stop early with given probability p

Converting Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo to Las Vegas

- Requires certificate to check if solution is correct
- If not correct, run Monte Carlo Algorithm again

Las Vegas to Monte Carlo

- Start Las Vegas algorithm and let run for set period of time
- Will stop early with given probability p
 - Solution guaranteed to be correct

Converting Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo to Las Vegas

- Requires certificate to check if solution is correct
- If not correct, run Monte Carlo Algorithm again

Las Vegas to Monte Carlo

- Start Las Vegas algorithm and let run for set period of time
- Will stop early with given probability p
 - Solution guaranteed to be correct
 - Return correct answer

Converting Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo to Las Vegas

- Requires certificate to check if solution is correct
- If not correct, run Monte Carlo Algorithm again

Las Vegas to Monte Carlo

- Start Las Vegas algorithm and let run for set period of time
- Will stop early with given probability p
 - Solution guaranteed to be correct
 - Return correct answer
- Will not stop early with probability $1 - p$

Converting Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo to Las Vegas

- Requires certificate to check if solution is correct
- If not correct, run Monte Carlo Algorithm again

Las Vegas to Monte Carlo

- Start Las Vegas algorithm and let run for set period of time
- Will stop early with given probability p
 - Solution guaranteed to be correct
 - Return correct answer
- Will not stop early with probability $1 - p$
 - Halt algorithm

Converting Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo to Las Vegas

- Requires certificate to check if solution is correct
- If not correct, run Monte Carlo Algorithm again

Las Vegas to Monte Carlo

- Start Las Vegas algorithm and let run for set period of time
- Will stop early with given probability p
 - Solution guaranteed to be correct
 - Return correct answer
- Will not stop early with probability $1 - p$
 - Halt algorithm
 - Return some answer

Converting Algorithms

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Monte Carlo to Las Vegas

- Requires certificate to check if solution is correct
- If not correct, run Monte Carlo Algorithm again

Las Vegas to Monte Carlo

- Start Las Vegas algorithm and let run for set period of time
- Will stop early with given probability p
 - Solution guaranteed to be correct
 - Return correct answer
- Will not stop early with probability $1 - p$
 - Halt algorithm
 - Return some answer
 - Need not be correct

Problems of Interest in Symbolic Linear Algebra

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Solving a Linear System of Equations

Problems of Interest in Symbolic Linear Algebra

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Solving a Linear System of Equations

- Solve $Ax = b$ where $A \in \mathbb{F}^{n \times n}$ and $b \in \mathbb{F}^n$

Problems of Interest in Symbolic Linear Algebra

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Solving a Linear System of Equations

- Solve $Ax = b$ where $A \in \mathbb{F}^{n \times n}$ and $b \in \mathbb{F}^n$
- A nonsingular or singular

Problems of Interest in Symbolic Linear Algebra

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Solving a Linear System of Equations

- Solve $Ax = b$ where $A \in \mathbb{F}^{n \times n}$ and $b \in \mathbb{F}^n$
- A nonsingular or singular

Matrix Determinant

Problems of Interest in Symbolic Linear Algebra

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Solving a Linear System of Equations

- Solve $Ax = b$ where $A \in \mathbb{F}^{n \times n}$ and $b \in \mathbb{F}^n$
- A nonsingular or singular

Matrix Determinant

- $\det(A)$ where $A \in \mathbb{F}^{n \times n}$

Problems of Interest in Symbolic Linear Algebra

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Solving a Linear System of Equations

- Solve $Ax = b$ where $A \in \mathbb{F}^{n \times n}$ and $b \in \mathbb{F}^n$
- A nonsingular or singular

Matrix Determinant

- $\det(A)$ where $A \in \mathbb{F}^{n \times n}$
- A nonsingular

Problems of Interest in Symbolic Linear Algebra

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Solving a Linear System of Equations

- Solve $Ax = b$ where $A \in \mathbb{F}^{n \times n}$ and $b \in \mathbb{F}^n$
- A nonsingular or singular

Matrix Determinant

- $\det(A)$ where $A \in \mathbb{F}^{n \times n}$
- A nonsingular

Matrix Rank

Problems of Interest in Symbolic Linear Algebra

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Solving a Linear System of Equations

- Solve $Ax = b$ where $A \in \mathbb{F}^{n \times n}$ and $b \in \mathbb{F}^n$
- A nonsingular or singular

Matrix Determinant

- $\det(A)$ where $A \in \mathbb{F}^{n \times n}$
- A nonsingular

Matrix Rank

- $\text{rank}(A)$ where $A \in \mathbb{F}^{n \times n}$

Problems of Interest in Symbolic Linear Algebra

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Solving a Linear System of Equations

- Solve $Ax = b$ where $A \in \mathbb{F}^{n \times n}$ and $b \in \mathbb{F}^n$
- A nonsingular or singular

Matrix Determinant

- $\det(A)$ where $A \in \mathbb{F}^{n \times n}$
- A nonsingular

Matrix Rank

- $\text{rank}(A)$ where $A \in \mathbb{F}^{n \times n}$
- A singular

Solving Linear Systems

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Nonsingular system)

$$\begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 3 & 3 & -2 & 3 \\ 0 & -3 & 0 & -2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 2 \\ 6 \\ -4 \end{bmatrix}$$

Gaussian Elimination

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example

$$\begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & -6 \\ 0 & 0 & 0 & -2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 2 \\ 6 \\ 2 \end{bmatrix} \implies x = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$

Gaussian Elimination

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example

$$\begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & -6 \\ 0 & 0 & 0 & -2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 2 \\ 6 \\ 2 \end{bmatrix} \implies x = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$

Gaussian Elimination

Gaussian Elimination

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example

$$\begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & -6 \\ 0 & 0 & 0 & -2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 2 \\ 6 \\ 2 \end{bmatrix} \implies x = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$

Gaussian Elimination

- Row echelon form and back substitution

Gaussian Elimination

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example

$$\begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & -6 \\ 0 & 0 & 0 & -2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 2 \\ 6 \\ 2 \end{bmatrix} \implies x = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$

Gaussian Elimination

- Row echelon form and back substitution
- Must know how matrix stored

Gaussian Elimination

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example

$$\begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & -6 \\ 0 & 0 & 0 & -2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 2 \\ 6 \\ 2 \end{bmatrix} \implies x = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$

Gaussian Elimination

- Row echelon form and back substitution
- Must know how matrix stored
- Matrix changed in calculation

Gaussian Elimination

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example

$$\begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & -6 \\ 0 & 0 & 0 & -2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 2 \\ 6 \\ 2 \end{bmatrix} \implies x = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$

Gaussian Elimination

- Row echelon form and back substitution
- Must know how matrix stored
- Matrix changed in calculation
- Sparse matrices may become dense

Black Box Matrix Model



Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Black Box Matrix Model



Black Box Matrix Model

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Black Box Matrix Model



Black Box Matrix Model

- External view of matrix

Black Box Matrix Model



Black Box Matrix Model

- External view of matrix
- Only matrix-vector products allowed

Black Box Matrix Model



Black Box Matrix Model

- External view of matrix
- Only matrix-vector products allowed
- Independent of implementation

Black Box Matrix Model



Black Box Matrix Model

- External view of matrix
- Only matrix-vector products allowed
- Independent of implementation
- Implementations may be efficient in time or space

Black Box Matrix Model



Black Box Matrix Model

- External view of matrix
- Only matrix-vector products allowed
- Independent of implementation
- Implementations may be efficient in time or space
- Not unique to symbolic computation

Black Box Matrix Model



Black Box Matrix Model

- External view of matrix
- Only matrix-vector products allowed
- Independent of implementation
- Implementations may be efficient in time or space
- Not unique to symbolic computation
- Can compute Krylov sequence $(A^i v)_{i \in \mathbb{Z}_{\geq 0}}$

Possible Black Box Matrices

Arbitrary Matrix

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots & a_{2,n} \\ a_{3,1} & a_{3,2} & a_{3,3} & \cdots & a_{3,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & a_{n,3} & \cdots & a_{n,n} \end{bmatrix}$$

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Possible Black Box Matrices

Arbitrary Matrix

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots & a_{2,n} \\ a_{3,1} & a_{3,2} & a_{3,3} & \cdots & a_{3,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & a_{n,3} & \cdots & a_{n,n} \end{bmatrix}$$

Storage: n^2 (store every entry)

Possible Black Box Matrices

Arbitrary Matrix

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots & a_{2,n} \\ a_{3,1} & a_{3,2} & a_{3,3} & \cdots & a_{3,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & a_{n,3} & \cdots & a_{n,n} \end{bmatrix}$$

Storage: n^2 (store every entry)

Time: $O(n^2)$

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Possible Black Box Matrices

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Arbitrary Matrix

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots & a_{2,n} \\ a_{3,1} & a_{3,2} & a_{3,3} & \cdots & a_{3,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & a_{n,3} & \cdots & a_{n,n} \end{bmatrix}$$

Storage: n^2 (store every entry)

Time: $O(n^2)$

Sparse Matrix

Only η nonzero entries

Possible Black Box Matrices

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Arbitrary Matrix

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots & a_{2,n} \\ a_{3,1} & a_{3,2} & a_{3,3} & \cdots & a_{3,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & a_{n,3} & \cdots & a_{n,n} \end{bmatrix}$$

Storage: n^2 (store every entry)

Time: $O(n^2)$

Sparse Matrix

Only η nonzero entries

Storage: $O(\eta)$ (store nonzero entries and indices)

Possible Black Box Matrices

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Arbitrary Matrix

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots & a_{2,n} \\ a_{3,1} & a_{3,2} & a_{3,3} & \cdots & a_{3,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & a_{n,3} & \cdots & a_{n,n} \end{bmatrix}$$

Storage: n^2 (store every entry)

Time: $O(n^2)$

Sparse Matrix

Only η nonzero entries

Storage: $O(\eta)$ (store nonzero entries and indices)

Time: $O(\eta)$

Possible Black Box Matrices

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Hilbert Matrix

$$A = \begin{bmatrix} \frac{1}{1} & \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{n} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots & \frac{1}{n+1} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots & \frac{1}{n+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{1}{n} & \frac{1}{n+1} & \frac{1}{n+1} & \cdots & \frac{1}{2n} \end{bmatrix}$$

Possible Black Box Matrices

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Hilbert Matrix

$$A = \begin{bmatrix} \frac{1}{1} & \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{n} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots & \frac{1}{n+1} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots & \frac{1}{n+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{1}{n} & \frac{1}{n+1} & \frac{1}{n+1} & \cdots & \frac{1}{2n} \end{bmatrix}$$

Storage: $O(1)$ (store only dimension n)

Possible Black Box Matrices

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Hilbert Matrix

$$A = \begin{bmatrix} \frac{1}{1} & \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{n} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots & \frac{1}{n+1} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots & \frac{1}{n+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{1}{n} & \frac{1}{n+1} & \frac{1}{n+1} & \cdots & \frac{1}{2n} \end{bmatrix}$$

Storage: $O(1)$ (store only dimension n)

Time: $O(n)$

Possible Black Box Matrices

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Toeplitz Matrix

$$A = \begin{bmatrix} a_n & a_{n+1} & a_{n+2} & \cdots & a_{2n-1} \\ a_{n-1} & a_n & a_{n+1} & \cdots & a_{2n-2} \\ a_{n-2} & a_{n-1} & a_n & \cdots & a_{2n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & a_3 & \cdots & a_n \end{bmatrix}$$

Possible Black Box Matrices

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Toeplitz Matrix

$$A = \begin{bmatrix} a_n & a_{n+1} & a_{n+2} & \cdots & a_{2n-1} \\ a_{n-1} & a_n & a_{n+1} & \cdots & a_{2n-2} \\ a_{n-2} & a_{n-1} & a_n & \cdots & a_{2n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & a_3 & \cdots & a_n \end{bmatrix}$$

Storage: $O(n)$

Possible Black Box Matrices

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Toeplitz Matrix

$$A = \begin{bmatrix} a_n & a_{n+1} & a_{n+2} & \cdots & a_{2n-1} \\ a_{n-1} & a_n & a_{n+1} & \cdots & a_{2n-2} \\ a_{n-2} & a_{n-1} & a_n & \cdots & a_{2n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & a_3 & \cdots & a_n \end{bmatrix}$$

Storage: $O(n)$

Time: $O(n \log(n) \log \log(n))$ (implement via fast
polynomial multiplication)

Linearly Recurrent Sequences

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Definition

Linearly Recurrent Sequences

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Definition

- Let $\mathbb{V}$ be a vector space over field $\mathbb{F}$.

Linearly Recurrent Sequences

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Definition

- Let $\mathbb{V}$ be a vector space over field $\mathbb{F}$.
- A sequence

$$a = (a_i)_{i \in \mathbb{Z}_{\geq 0}} \in \mathbb{V}^{\mathbb{Z}_{\geq 0}}$$

is *linearly recurrent* if and only if there exist $m \in \mathbb{Z}_{\geq 0}$ and

$$c_0, \dots, c_m \in \mathbb{F}, \quad c_m \neq 0$$

such that for all $i \geq 0$

Linearly Recurrent Sequences

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Definition

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such that for all $i \geq 0$

$$\sum_{j=0}^m c_j a_{i+j} = 0$$

Linearly Recurrent Sequences

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Definition

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$$c_0, \dots, c_m \in \mathbb{F}, \quad c_m \neq 0$$

such that for all $i \geq 0$

$$a_{i+m} = -\frac{1}{c_m} \left(\sum_{j=0}^{m-1} c_j a_{i+j} \right)$$

Linearly Recurrent Sequences

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

Linearly Recurrent Sequences

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

- $(a_i)_{i \in \mathbb{Z}_{\geq 0}} = (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$

Linearly Recurrent Sequences

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

- $(a_i)_{i \in \mathbb{Z}_{\geq 0}} = (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$
- $\mathbb{V} = \mathbb{F} = \mathbb{R}$

Linearly Recurrent Sequences

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

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Linearly Recurrent Sequences

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

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- $a_{i+2} - a_{i+1} - a_i = 0$

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Definition

The polynomial $f(\lambda) = \sum_{j=0}^m c_j \lambda^j$ generates a

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Definition

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Example (Fibonacci Numbers)

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Definition

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Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Definition

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Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

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The polynomial $f(\lambda) = \sum_{j=0}^m c_j \lambda^j$ generates a

Example (Fibonacci Numbers)

- $(a_i)_{i \in \mathbb{Z}_{\geq 0}} = (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$
- $a_{i+2} - a_{i+1} - a_i = 0$
- $f = \lambda^2 - \lambda - 1$ generates the Fibonacci sequence a

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Module

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Module

- Define $f \bullet a = \left(\sum_{j=0}^m c_j a_{i+j} \right)_{i \in \mathbb{Z}_{\geq 0}} = (0)_{i \in \mathbb{Z}_{\geq 0}} = 0 \in \mathbb{V}^{\mathbb{Z}_{\geq 0}}$

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Module

- Define $f \bullet a = \left(\sum_{j=0}^m c_j a_{i+j} \right)_{i \in \mathbb{Z}_{\geq 0}} = (0)_{i \in \mathbb{Z}_{\geq 0}} = 0 \in \mathbb{V}^{\mathbb{Z}_{\geq 0}}$
- $\mathbb{V}^{\mathbb{Z}_{\geq 0}}$ is an $\mathbb{F}[\lambda]$ -module

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Module

- Define $f \bullet a = \left(\sum_{j=0}^m c_j a_{i+j} \right)_{i \in \mathbb{Z}_{\geq 0}} = (0)_{i \in \mathbb{Z}_{\geq 0}} = 0 \in \mathbb{V}^{\mathbb{Z}_{\geq 0}}$
- $\mathbb{V}^{\mathbb{Z}_{\geq 0}}$ is an $\mathbb{F}[\lambda]$ -module
- If $f \bullet a = 0$ and $g \in \mathbb{F}[\lambda]$, then

$$(g f) \bullet a = g \bullet (f \bullet a) = g \bullet 0 = 0$$

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

- $(a_i)_{i \in \mathbb{Z}_{\geq 0}} = (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

- $(a_i)_{i \in \mathbb{Z}_{\geq 0}} = (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$
- $f = \lambda^2 - \lambda - 1$ generates the Fibonacci sequence a

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

- $(a_i)_{i \in \mathbb{Z}_{\geq 0}} = (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$
- $f = \lambda^2 - \lambda - 1$ generates the Fibonacci sequence a
- $(\lambda + 1)f = \lambda^3 - 2\lambda - 1$ also generates a

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

- $(a_i)_{i \in \mathbb{Z}_{\geq 0}} = (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$
- $f = \lambda^2 - \lambda - 1$ generates the Fibonacci sequence a
- $(\lambda + 1)f = \lambda^3 - 2\lambda - 1$ also generates a
 - $a_{i+3} - 2a_{i+1} - a_i = 0$

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

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- $f = \lambda^2 - \lambda - 1$ generates the Fibonacci sequence a
- $(\lambda + 1)f = \lambda^3 - 2\lambda - 1$ also generates a
 - $a_{i+3} - 2a_{i+1} - a_i = 0$
 - $a_{i+3} = 2a_{i+1} + a_i$

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

- $(a_i)_{i \in \mathbb{Z}_{\geq 0}} = (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$
- $f = \lambda^2 - \lambda - 1$ generates the Fibonacci sequence a
- $(\lambda + 1)f = \lambda^3 - 2\lambda - 1$ also generates a
 - $a_{i+3} - 2a_{i+1} - a_i = 0$
 - $a_{i+3} = 2a_{i+1} + a_i$
- $\lambda^k f = \lambda^{k+2} - \lambda^{k+1} - \lambda^k$ also generates a

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

- $(a_i)_{i \in \mathbb{Z}_{\geq 0}} = (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$
- $f = \lambda^2 - \lambda - 1$ generates the Fibonacci sequence a
- $(\lambda + 1)f = \lambda^3 - 2\lambda - 1$ also generates a
 - $a_{i+3} - 2a_{i+1} - a_i = 0$
 - $a_{i+3} = 2a_{i+1} + a_i$
- $\lambda^k f = \lambda^{k+2} - \lambda^{k+1} - \lambda^k$ also generates a
 - $a_{i+k+2} - a_{i+k+1} - a_{i+k} = 0$

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

- $(a_i)_{i \in \mathbb{Z}_{\geq 0}} = (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$
- $f = \lambda^2 - \lambda - 1$ generates the Fibonacci sequence a
- $(\lambda + 1)f = \lambda^3 - 2\lambda - 1$ also generates a
 - $a_{i+3} - 2a_{i+1} - a_i = 0$
 - $a_{i+3} = 2a_{i+1} + a_i$
- $\lambda^k f = \lambda^{k+2} - \lambda^{k+1} - \lambda^k$ also generates a
 - $a_{i+k+2} - a_{i+k+1} - a_{i+k} = 0$
 - $a_{i+k+2} = a_{i+k+1} + a_{i+k}$

Generating Polynomials

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example (Fibonacci Numbers)

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 - $a_{i+k+2} - a_{i+k+1} - a_{i+k} = 0$
 - $a_{i+k+2} = a_{i+k+1} + a_{i+k}$
 - Skips first k elements of a

Minimal Generating Polynomial

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Existence

Minimal Generating Polynomial

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Existence

- $\{f \in \mathbb{F}[\lambda] \mid f \bullet a = 0\}$ is an ideal

Minimal Generating Polynomial

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Existence

- $\{f \in \mathbb{F}[\lambda] \mid f \bullet a = 0\}$ is an ideal
- $\mathbb{F}[\lambda]$ is a principal ideal domain

Minimal Generating Polynomial

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Existence

- $\{f \in \mathbb{F}[\lambda] \mid f \bullet a = 0\}$ is an ideal
- $\mathbb{F}[\lambda]$ is a principal ideal domain
- There exists a unique monic generator of minimal degree, the *minimal generating polynomial* of sequence

Minimal Generating Polynomial

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Existence

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- Minimal polynomial divides all generating polynomials

Minimal Generating Polynomial

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Existence

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- There exists a unique monic generator of minimal degree, the *minimal generating polynomial* of sequence
- Minimal polynomial divides all generating polynomials

Example (Fibonacci Numbers)

The polynomial $f = \lambda^2 - \lambda - 1$ is the minimal generating polynomial of the Fibonacci sequence a .

Matrix Power Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Matrix Power Sequence

Matrix Power Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Matrix Power Sequence

- $f \bullet (A^i)_{i \in \mathbb{Z}_{\geq 0}}$ if and only if $f(A) = 0$

Matrix Power Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Matrix Power Sequence

- $f \bullet (A^i)_{i \in \mathbb{Z}_{\geq 0}}$ if and only if $f(A) = 0$
- $\det(\lambda I - A)$ generates $(A^i)_{i \in \mathbb{Z}_{\geq 0}}$

Matrix Power Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Matrix Power Sequence

- $f \bullet (A^i)_{i \in \mathbb{Z}_{\geq 0}}$ if and only if $f(A) = 0$
- $\det(\lambda I - A)$ generates $(A^i)_{i \in \mathbb{Z}_{\geq 0}}$
 - Cayley-Hamilton Theorem

Matrix Power Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Matrix Power Sequence

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- $\det(\lambda I - A)$ generates $(A^i)_{i \in \mathbb{Z}_{\geq 0}}$
 - Cayley-Hamilton Theorem
- Let f^A be the minimal polynomial of $(A^i)_{i \in \mathbb{Z}_{\geq 0}}$.

Matrix Power Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Matrix Power Sequence

- $f \bullet (A^i)_{i \in \mathbb{Z}_{\geq 0}}$ if and only if $f(A) = 0$
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 - $f^A \mid \det(\lambda I - A)$

Matrix Power Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Matrix Power Sequence

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 - $f^A \mid \det(\lambda I - A)$
 - $\deg(f^A) \leq \deg(\det(\lambda I - A)) = n$

Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Krylov Sequence

Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Krylov Sequence

- $f \bullet (A^i v)_{i \in \mathbb{Z}_{\geq 0}}$ if and only if $f(A) v = 0$

Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Krylov Sequence

- $f \bullet (A^i v)_{i \in \mathbb{Z}_{\geq 0}}$ if and only if $f(A) v = 0$
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Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Krylov Sequence

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Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Krylov Sequence

- $f \bullet (A^i v)_{i \in \mathbb{Z}_{\geq 0}}$ if and only if $f(A) v = 0$
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- Let $f^{A,v}$ be the minimal polynomial of $(A^i v)_{i \in \mathbb{Z}_{\geq 0}}$.

Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Krylov Sequence

- $f \bullet (A^i v)_{i \in \mathbb{Z}_{\geq 0}}$ if and only if $f(A) v = 0$
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- Let $f^{A,v}$ be the minimal polynomial of $(A^i v)_{i \in \mathbb{Z}_{\geq 0}}$.
 - $f^{A,v} \mid f^A \mid \det(\lambda I - A)$

Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Krylov Sequence

- $f \bullet (A^i v)_{i \in \mathbb{Z}_{\geq 0}}$ if and only if $f(A) v = 0$
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 - $\deg(f^{A,v}) \leq \deg(f^A) \leq \deg(\det(\lambda I - A)) = n$

Solving a Linear System

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Using Linear Recurrences to Solve a Linear System

Solving a Linear System

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Using Linear Recurrences to Solve a Linear System

- Suppose $g = \sum_{j=0}^m c_j \lambda^j$, $g(0) = c_0 \neq 0$, and $g \bullet (A^i b)_{i \in \mathbb{Z}_{\geq 0}}$

Solving a Linear System

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Using Linear Recurrences to Solve a Linear System

- Suppose $g = \sum_{j=0}^m c_j \lambda^j$, $g(0) = c_0 \neq 0$, and $g \bullet (A^i b)_{i \in \mathbb{Z}_{\geq 0}}$
- If g exists, then $f^{A,b}$ satisfies the requirements.

Solving a Linear System

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Using Linear Recurrences to Solve a Linear System

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- If g exists, then $f^{A,b}$ satisfies the requirements.
- Consider linear combination
$$c_0 b + c_1 A b + \cdots + c_m A^m b = 0$$

Solving a Linear System

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Using Linear Recurrences to Solve a Linear System

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- If g exists, then $f^{A,b}$ satisfies the requirements.
- Consider linear combination
$$c_0 b + c_1 A b + \cdots + c_m A^m b = 0$$
- $$b = -\frac{1}{c_0} \sum_{j=1}^m (A^j b) = A \left(-\frac{1}{c_0} \sum_{j=1}^m (c_j A^{j-1} b) \right)$$

Solving a Linear System

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Using Linear Recurrences to Solve a Linear System

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- If g exists, then $f^{A,b}$ satisfies the requirements.
- Consider linear combination
$$c_0 b + c_1 A b + \cdots + c_m A^m b = 0$$
- $$b = -\frac{1}{c_0} \sum_{j=1}^m (A^j b) = A \left(-\frac{1}{c_0} \sum_{j=1}^m (c_j A^{j-1} b) \right)$$
- $$x = -\frac{1}{c_0} \sum_{j=1}^m (c_j A^{j-1} b)$$
 is a solution

Nonsingular Matrix

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Nonsingular A

Nonsingular Matrix

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Nonsingular A

- $\det(A) \neq 0$

Nonsingular Matrix

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Nonsingular A

- $\det(A) \neq 0$
- $\det(\lambda I - A)|_{\lambda=0} \neq 0$

Nonsingular Matrix

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Nonsingular A

- $\det(A) \neq 0$
- $\det(\lambda I - A)|_{\lambda=0} \neq 0$
- $f^{A,b}(0) \neq 0$

Nonsingular Matrix

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Nonsingular A

- $\det(A) \neq 0$
- $\det(\lambda I - A)|_{\lambda=0} \neq 0$
- $f^{A,b}(0) \neq 0$
- $x = -\frac{1}{c_0} \sum_{j=1}^m (c_j A^{j-1} b)$ is the unique solution

Computing the Minimal Polynomial of Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Bilinear Projection Sequence

Computing the Minimal Polynomial of Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Bilinear Projection Sequence

- $f \bullet (u^T A^i v)_{i \in \mathbb{Z}_{\geq 0}}$ if and only if $u^T f(A) v = 0$

Computing the Minimal Polynomial of Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Bilinear Projection Sequence

- $f \bullet (u^T A^i v)_{i \in \mathbb{Z}_{\geq 0}}$ if and only if $u^T f(A) v = 0$
- $f(A) v = 0$ means $u^T f(A) v = 0$

Computing the Minimal Polynomial of Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Bilinear Projection Sequence

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Computing the Minimal Polynomial of Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Bilinear Projection Sequence

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- Let $f_u^{A,v}$ be the minimal polynomial of $(u^T A^i v)_{i \in \mathbb{Z}_{\geq 0}}$.

Computing the Minimal Polynomial of Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Bilinear Projection Sequence

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 - $f_u^{A,v} \mid f^{A,v} \mid f^A \mid \det(\lambda I - A)$

Computing the Minimal Polynomial of Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Bilinear Projection Sequence

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 - $f_u^{A,v} \mid f^{A,v} \mid f^A \mid \det(\lambda I - A)$
 - $\deg(f_u^{A,v}) \leq \deg(f^{A,v}) \leq \deg(f^A) \leq \deg(\det(\lambda I - A)) = n$

Computing the Minimal Polynomial of Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Minimal Polynomials Probably Equal

If u chosen randomly from finite $S^n \subset \mathbb{F}^n$, then $f_u^{A,v} = f^{A,v}$ with probability at least

$$1 - \frac{\deg(f^{A,v})}{|S|}$$

Computing the Minimal Polynomial of Krylov Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

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If u chosen randomly from finite $S^n \subset \mathbb{F}^n$, then $f_u^{A,v} = f^{A,v}$ with probability at least

$$1 - \frac{\deg(f^{A,v})}{|S|}$$

Certificate

$$f_u^{A,v} \bullet (A^i v)_{i \in \mathbb{Z}_{\geq 0}} = 0$$

Computing the Minimal Polynomial of Scalar Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bound

Must know bound $M \geq \deg(f)$

Computing the Minimal Polynomial of Scalar Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bound

Must know bound $M \geq \deg(f)$

Extended Euclidean Algorithm

Computing the Minimal Polynomial of Scalar Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bound

Must know bound $M \geq \deg(f)$

Extended Euclidean Algorithm

- $s_j f_{-1} + t_j f_0 = f_j$

Computing the Minimal Polynomial of Scalar Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bound

Must know bound $M \geq \deg(f)$

Extended Euclidean Algorithm

- $s_j f_{-1} + t_j f_0 = f_j$
- Inputs are $f_{-1} = \lambda^{2M}$ and $f_0 = \sum_{i=0}^{2M-1} (a_i \lambda^i)$

Computing the Minimal Polynomial of Scalar Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bound

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- Inputs are $f_{-1} = \lambda^{2M}$ and $f_0 = \sum_{i=0}^{2M-1} (a_i \lambda^i)$
- Stop when $\deg(f_j) \leq M < \deg(f_{j-1})$

Computing the Minimal Polynomial of Scalar Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bound

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- Stop when $\deg(f_j) \leq M < \deg(f_{j-1})$
- Minimal polynomial is reversal of $t_j(\lambda)$

Computing the Minimal Polynomial of Scalar Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bound

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- Stop when $\deg(f_j) \leq M < \deg(f_{j-1})$
- Minimal polynomial is reversal of $t_j(\lambda)$

Berlekamp-Massey Algorithm

Computing the Minimal Polynomial of Scalar Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bound

Must know bound $M \geq \deg(f)$

Extended Euclidean Algorithm

- $s_j f_{-1} + t_j f_0 = f_j$
- Inputs are $f_{-1} = \lambda^{2M}$ and $f_0 = \sum_{i=0}^{2M-1} (a_i \lambda^i)$
- Stop when $\deg(f_j) \leq M < \deg(f_{j-1})$
- Minimal polynomial is reversal of $t_j(\lambda)$

Berlekamp-Massey Algorithm

- From coding theory

Computing the Minimal Polynomial of Scalar Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bound

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- Interpolates elements of scalar sequence

Computing the Minimal Polynomial of Scalar Sequence

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bound

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Berlekamp-Massey Algorithm

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- Related to Extended Euclidean Algorithm

Wiedemann's Algorithm

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background

The Method

Degree Bounds

Wiedemann's Algorithm

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bounds

- $f_u^{A,v} \mid f^{A,v} \mid f^A \mid \det(\lambda I - A)$

Wiedemann's Algorithm

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bounds

- $f_u^{A,v} \mid f^{A,v} \mid f^A \mid \det(\lambda I - A)$
- $\deg(f_u^{A,v}) \leq \deg(\det(\lambda I - A)) = n$

Wiedemann's Algorithm

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bounds

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- Only need $(u^T A^i b)_{i=0}^{2n-1}$

Wiedemann's Algorithm

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bounds

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Wiedemann's Algorithm

Wiedemann's Algorithm

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bounds

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Wiedemann's Algorithm

Require: nonsingular $A \in \mathbb{F}^{n \times n}$ and $b \in \mathbb{F}^n$

Wiedemann's Algorithm

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bounds

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Ensure: $x \in \mathbb{F}^n$ such that $A x = b$

Wiedemann's Algorithm

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bounds

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Ensure: $x \in \mathbb{F}^n$ such that $A x = b$

1: $u \leftarrow$ random vector in S^n where $S \subset \mathbb{F}$

Wiedemann's Algorithm

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bounds

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Ensure: $x \in \mathbb{F}^n$ such that $A x = b$

- 1: $u \leftarrow$ random vector in S^n where $S \subset \mathbb{F}$
- 2: use Berlekamp-Massey to compute
 $f^{A,b} = c_0 + c_1 \lambda + \cdots + c_m \lambda^m$ {Store only $A^i b$ }

Wiedemann's Algorithm

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Degree Bounds

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Ensure: $x \in \mathbb{F}^n$ such that $Ax = b$

- 1: $u \leftarrow$ random vector in S^n where $S \subset \mathbb{F}$
- 2: use Berlekamp-Massey to compute
 $f^{A,b} = c_0 + c_1\lambda + \cdots + c_m\lambda^m$ {Store only $A^i b$ }
- 3: $x \leftarrow -\frac{1}{c_0}(c_1 b + c_2 A b + \cdots + c_m A^{m-1} b)$

Back to original problem

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background

The Method

Example

Back to original problem

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example

$$\begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 3 & 3 & -2 & 3 \\ 0 & -3 & 0 & -2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 2 \\ 6 \\ -4 \end{bmatrix}$$

Back to original problem

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example

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- $n = 4$

Back to original problem

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example

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- $n = 4$

- $u^T = [2 \ 1 \ 1 \ 2]$

Back to original problem

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- $\begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 3 & 3 & -2 & 3 \\ 0 & -3 & 0 & -2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 2 \\ 6 \\ -4 \end{bmatrix}$
- $n = 4$
- $u^T = [2 \ 1 \ 1 \ 2]$
- $(u^T A^i b)_{i=0}^{2n-1} = (8, -4, 20, -28, 68, -124, 260, -508)$

Back to original problem

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example

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Back to original problem

Linear Algebra
on a
Computer

W. J. Turner

Symbolic
Computation

Symbolic
Algorithms

Black Box
Matrix Model

Wiedemann
Method

Background
The Method

Example

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- $(u^T A^i b)_{i=0}^{2n-1} = (8, -4, 20, -28, 68, -124, 260, -508)$

- $f^{A,b} = f_u^{A,b} = \lambda^2 + \lambda - 2$

- $$x = \frac{1}{2} (Ab + b) = \frac{1}{2} \left(\begin{bmatrix} -2 \\ 2 \\ -6 \\ 2 \end{bmatrix} + \begin{bmatrix} 4 \\ 2 \\ 6 \\ -4 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$